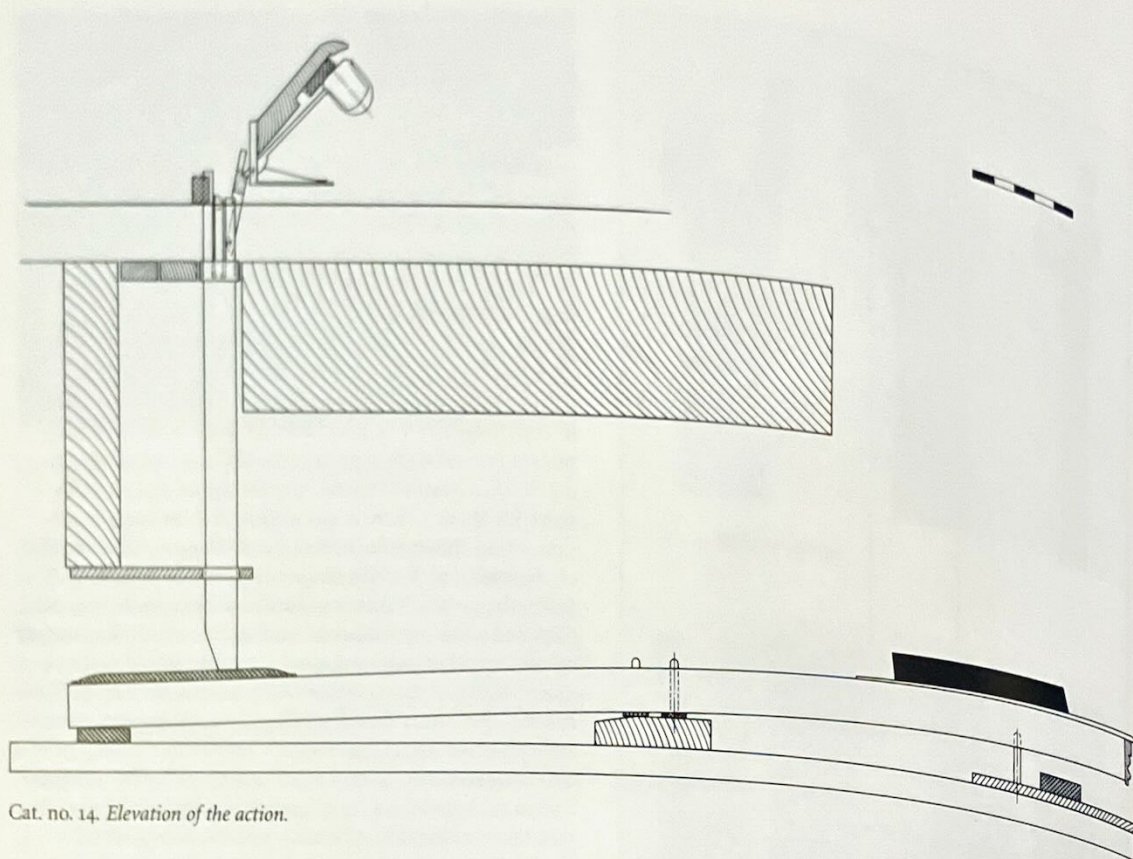




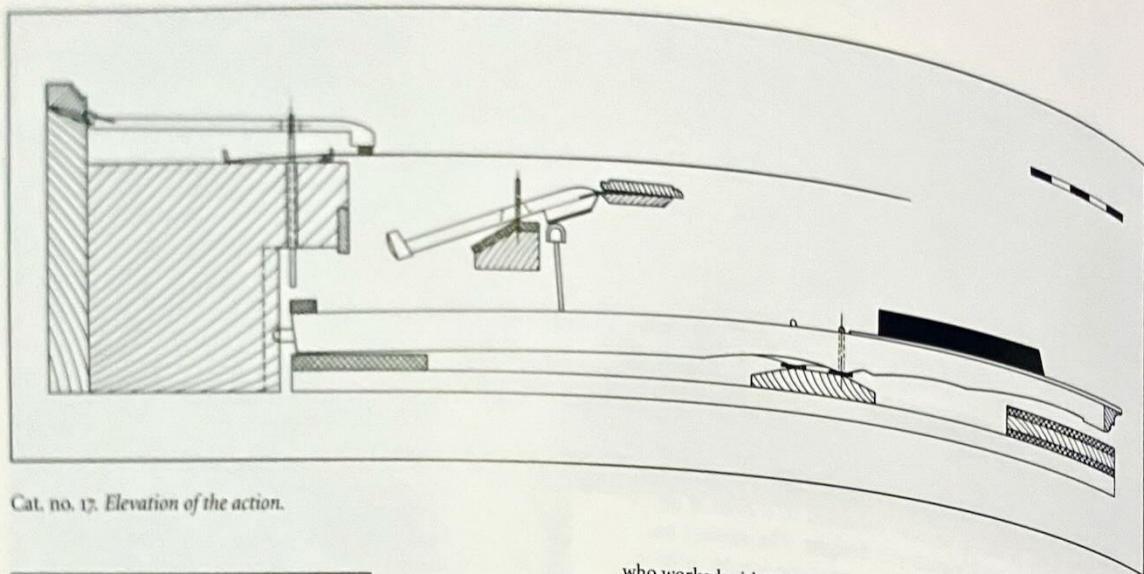
Cat. no. 14. Elevation of the action.

tledrums, harps, violins, other bowed strings,²¹ and "nail" violins.²² His keyboard instruments are, perhaps, of greater interest. Some of these, such as square,²³ grand,²⁴ and upright²⁵ pianos, and even a square piano combined with an organ²⁶ were relatively conventional. Others, such as a "vocal harp" (a keyboard instrument with bowed strings)²⁷ and, above all, the combined harpsichords and pianos, might appear to be musically useless tours de force. As was the case with Merlin's mechanical devices, however, his seemingly unconventional keyboard instruments were significant and useful in their day, and, in fact, not all that unusual.²⁸

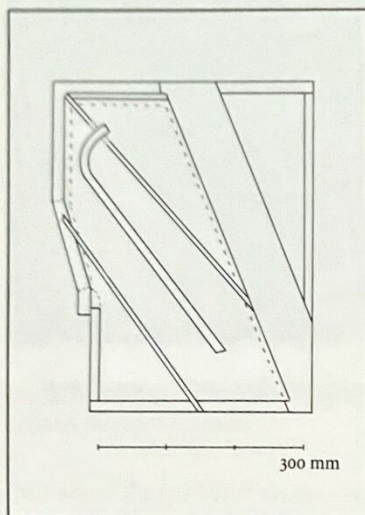
When in 1711 Scipione Maffei described the piano invented by Bartolomeo Cristofori, it was called *gravecembalo col piano e forte*, i.e., "harpsichord [capable of sounding] soft and loud."²⁹ Cristofori's instrument was a piano as the term is understood today, but the early terminology indicates that it was regarded as a type of harpsichord with an alternative action. That this view persisted throughout much of the eighteenth century is shown by other names for and descriptions of the piano. The texts include Merlin's own wording in his 1774 patent for "a new kind of compound harpsichord, in which, besides the jacks with quills, a set of hammers of the nature of those used in the kind of harpsichord called piano forte, are introduced."³⁰ The lack of a clear distinction between piano and harpsichord quite naturally led to efforts to combine the

two actions in a single instrument. Merlin's compound harpsichord had been preceded by almost sixty years of similar work by others,³¹ the earliest conception by Jean Marius of Paris in 1716.³² Others who anticipated Merlin were Giovanni Ferrini of Florence in 1746,³³ an unknown Hamburg maker whose instrument was brought to England in 1755 by the poet William Mason,³⁴ Andries Veltman in collaboration with a certain Dumontier in Paris in 1757–1759,³⁵ and Johann Andreas Stein of Augsburg by 1769.³⁶ These instruments varied considerably from one another³⁷ and from Merlin's version; thus, each design was, in some sense, original. Merlin's is distinctive in having a downstriking action that could rather easily be applied to older harpsichords. In addition to the Museum's 1758 Kirkman to which Merlin added the piano action, another of his instruments of this type, made in 1780, has survived.³⁸

Although the combination of harpsichord and piano action within a single instrument might seem strange and awkward, Merlin's were accepted by notable musicians of the period. In April 1774 Johann Christian Bach, the most important composer active in England at the time, seems to have introduced Merlin's instrument in two concerts at which he played "a new Concerto upon Mr Merlin's lately-invented harpsichord."³⁹ By 1775 the influential musicologist Charles Burney had purchased one.⁴⁰ The success of the concept is



Cat. no. 17. Elevation of the action.



Cat. no. 17. Plan of the soundboard.

(i.e., without stringing, crossbanding, or blackened moldings), the cartouche was merely glued to the underlying surface and remains raised above.

4. This was the form of Zumppe's name found in the archives of his birthplace; see *Nordbayerische Zeitung* 1957. (I am most grateful to John Henry van der Meer for having brought to my attention this important source for Zumppe's biography and for having provided me with a copy of it.) The best source of information about Zumppe's career in England is Cole 1986. See also James 1930, pp. 50–51, 80, and 137; and Grove 1984, s.v. "Zumppe, Johannes."

5. See *Nordbayerische Zeitung* 1957.

6. *Ibid.* There is no evidence for the often-repeated assertion that Zumppe worked for Silbermann. Neither Zumppe nor any of the English piano makers are mentioned in any of the hundreds of Silbermann documents collected in Müller 1982. (Only Frederick Schoene, Zumppe's successor, might have been related to the Saxon organ builder Johann Georg Schön,

who worked with Silbermann.) The supposed association of Zumppe with Silbermann probably stems from nationalistically tinged mid-nineteenth-century German sources such as Welcker von Gontershausen 1855, p. 120: "In 1764 a German named Zumppe [sic] brought the Silbermann *Stossmechanik* to London."

7. It is often stated (for example, in Grove 1984, s.v. "Zumppe, Johannes") that Zumppe and other German piano makers emigrated to London because of the disruptions caused in Germany by the Seven Years War (1756–1763). The Nuremberg area was not, however, one of those most directly affected by the war but had been gradually declining in prosperity since the sixteenth century. Thus, on the basis of Burney's statement that Zumppe had "long worked for Shudi," it seems most likely that Zumppe left home to seek his fortune abroad as a young journeyman carpenter, perhaps in the late 1740s when he would have been in his early 20s.

8. See Cole 1986, p. 564.

9. On modern and eighteenth-century maps (see, for example, Cole 1986, p. 564) it appears as "Princes." According to Maunder 1989, p. 84, however, the nameboards of Zumppe pianos of the 1760s show the spelling "Princess." The pianos of 1767 mentioned in note 1 do indeed read "Princess" (I am grateful to Laurence Libin for confirming the spelling on the New York instrument for me; see also Schott 1985, no. 29), but the MFA piano of 1770 and the instrument of 1776 mentioned in note 1 read "Princes."

10. The instrument is in the Historisches Museum, Frankfurt am Main; see Baines 1966, pls. 251 and 252. According to Hipkins 1929, p. 105, "[François-Joseph] Fétis [1784–1871] says he began his pianoforte studies on a Zumppe Square, dated 1762."

11. Three square pianos of this year survive: in Cambridge (see note 1); in the Württembergisches Landesmuseum in Stuttgart; and in the collection of William Garlick, Boston (see Schott 1985, no. 29).

12. J. C. Bach came to London in 1762, and his first concert in conjunction with Abel was given in January 1765 (see Grove 1980, s.v. "Bach . . . Johann Christian"). The frequently retold story of Bach having given in 1768 the first solo piano performance in England on a Zumppe instrument (see, for example, Grove 1984, s.v. "Zumppe, Johannes") has been shown to have been based on a misinterpretation of the documentary sources; see Cole 1986, pp. 563–564.

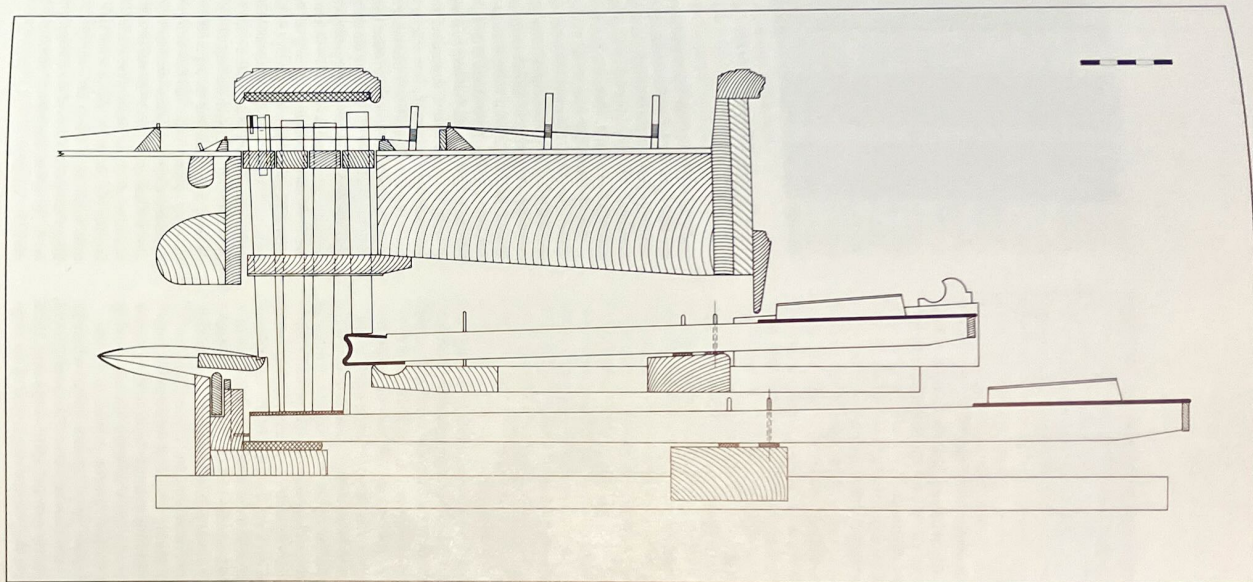
13. See Rees 1819, s.v. "Harpsichord."

A large industrial mold assembly, likely for casting, featuring numerous vertical channels and a complex mechanical structure. The mold is composed of many parallel, vertical, ribbed sections that form a series of channels. These sections are held together by a complex system of horizontal and vertical metal plates and bolts. The overall structure is rectangular and appears to be made of heavy-duty metal. The lighting is dramatic, with strong highlights and shadows, emphasizing the industrial and mechanical nature of the equipment.

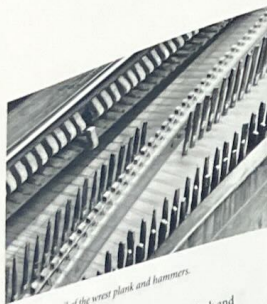
Cat. no. 21. Elevation of the action

plank are marks showing that there were small blocks to hold a music rack. A lock, covered by a plain rectangular brass plate, is at the front of the case. The instrument rests on four turned walnut screw-on legs terminating in brass casters. Each leg is decorated with six gilded flutes and with bronze studs above and below the flutes. Between the lower studs are small incised squares that, together with the studs, echo the egg-and-dart pattern on the molding on the case above.

In 1749 Stein made his way (perhaps with a detour to Vienna¹¹) to Regensburg, where he worked for several months



Cat. no. 7. Elevation of the action.



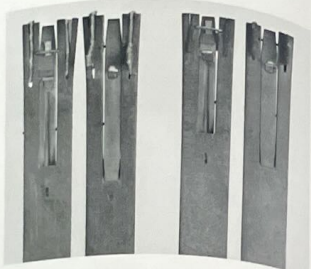
Cat. no. 14. Detail of the wrest plank and hammers.

jack rail and the case walls around the wrest plank and soundboard are veneered in walnut with its grain parallel to the spine. The soundboard is surrounded, at the case walls by a walnut molding that serves as the 8' hitch-pin rail. In the treble there is a half-round pearwood molding over the front edge of the soundboard. There is no music desk. The walnut lid and lid flaps are surrounded on their outer edges by a walnut molding. (The flap over the piano hammer rail is modified by Merlin to clear the piano hammer rail.) That the lid stick is of walnut— including the pedals, the cover over the mahogany (the latter with a holly the hammer rail—are of mahogany) includes three decorative cartouches. The brass hardware includes three for each of the lid flaps, and three S-shaped lid hooks on the cheekpiece and bent side. The walnut trestle stand, with baluster turnings terminating in casters with wooden rollers, can be disassembled by loosening the bed bolts into the stretchers.

THIS INSTRUMENT is of interest both as an example of a typical mid-eighteenth-century English single-manual harpsichord by a leading maker and, the more so, as a rare example of a combined harpsichord and piano. Further, its piano action is probably the earliest extant example of the down-pipe of a combined harpsichord and piano. Kirkman (1710–1791), was an Alsatian cabinetmaker of Swiss extraction who emigrated to England in the early 1730s.⁹ In London he worked for Hermann Tabel, a harpsichord maker from the low countries, who is said to have learned his craft in Antwerp with the successors to the Backerses.¹⁰ Shortly after Tabel's death in 1738, one morning at breakfast, Kirkman, who had been Tabel's foreman and executor, proposed marriage to his widow, provided that it take place before noon that day. She accepted, and Kirkman gained possession of his master's stock and business. Kirkman, a shrewd business-



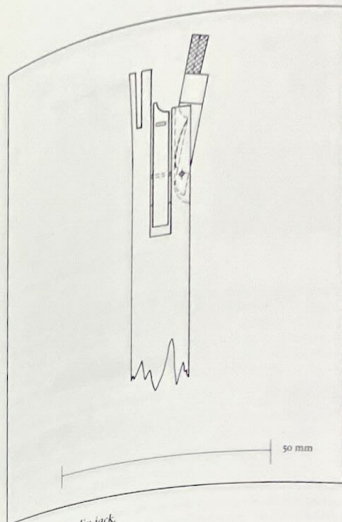
Cat. no. 14. Jacks by Merlin.



Cat. no. 14. Jacks by Kirkman, 4' (at the left), 8' (at the right).

man who amassed a large fortune that he is said to have increased by lending money at usurious rates, seems to have made almost three times the number of harpsichords made by his closest competitor, Burkat Shudi.¹¹

Most English harpsichords made after about 1740 are highly standardized in design. This design probably stemmed from Tabel, for whom Shudi had also worked. Generally the compass was FF to f'' and the disposition included two 8' sets of strings and one 4'.¹² Except for obvious additions—the second keyboard, the extension of the front of the case to accommodate it, and the provision of a fourth row of jacks (a nasal register on the upper manual)—two-manual instruments differ little from single-manual instruments, which have similar compasses, layouts, and scalings.¹³ A peculiarity of English harpsichords with 4' stops is that, for the upper



Cat. no. 14. Merlin jack.

three octaves or so, the 4' wrest pins were placed in front of the 8' nut, which was bored so that the 4' strings could pass through. This was done presumably because there would have been insufficient room for the 4' wrest pins in their usual position in this area. Such, apparently, was the standard-nasal register in this area. Such, apparently, was the standard-nasal register of design that even instruments without nasal registers—i.e., most single manual harpsichords—also have 4' wrest pins in front of the 8' nut.

Eighteenth-century English harpsichords were remarkably conservative in their decoration. The Museum's 1758 Kirkman was veneered with walnut about twenty-five years after mahogany had become fashionable; its decorative hinges and trestle stand with baluster turnings were more than fifty years out of style. It is likely that keyboard instruments, like clocks, which were often similarly old-fashioned, were regarded more as tools than as pieces of furniture to adorn a room.¹⁴

Harpsichords very much made in England until the end of the eighteenth century. Starting about 1760, there began to be improvements upon English harpsichords several optional "improvements" that reflected changes in musical taste leading to the

successful introduction of the piano into England in the same period. The first of these improvements to the harpsichord pedal that allowed the performer to turn registers of jacks off or on.¹⁵ Thus *crescendi* or *diminuendi* could be obtained without removing the hands from the keyboard to manipulate stops. Another improvement, patented by Burkat Shudi in 1769,¹⁶ was the Venetian swell such as is found on the Museum's Shudi and Broadwood harpsichord of 1772 (cat. no. 20). This device also allows the performer to control the volume with a pedal.

The "machine" was applied to the Museum's 1758 Kirkman in 1779, when a far more radical addition, the piano action, was also made by John Joseph Merlin (1735–1803; born in Huy, Belgium).¹⁷ In 1760, after six years in Paris, Merlin, trained in clockmaking or a similar skill, was taken to London by the Count of Fuentes, recently appointed an ambassador for Spain. (There is speculation that the count's mission was to obtain information about the newly developed chronometer, with which longitudes could be calculated.)¹⁸ In London, where Merlin remained until his death, he made, displayed in his own museum, and sold an astounding variety of mechanical devices, including automata, wheelchairs, toy tissier ovens, clocks, scales, moving models such as the "Circus of Cupid" and "Temple of Flora," a gambling machine, various musical instruments. Cultivating the persona of a charming eccentric, Merlin would appear at public masquerades in such guises as Cupid forging darts, which he would then throw at the ladies; Fortuna with her fateful wheel; a barmaid, complete with a bar; or a doctor in a wheelchair that gave an electric shock to anyone who touched it. Once, playing the violin while racing on roller skates, he "impelled himself against a mirror of more than five hundred pounds value, dashed it to atoms, broke his instrument to pieces and wounded himself most severely." Such escapades, along with witty speech in fractured English, helped to endear him to a circle of upper-class intellectuals that included Horace Walpole, Jeremy Bentham, Thomas Gainsborough (who painted Merlin's portrait), and Charles Burney, whose daughter Fanny, the novelist, recorded vivid impressions of Merlin in her diary.

It would be easy to dismiss many of Merlin's creations as amusing toys, but even the fanciful automata were of importance in that deistic era in which God was cast in the image of the clockmaker.¹⁹ Merlin's efforts may be seen as having something of the nature of modern experiments in artificial intelligence. There is, indeed, an almost direct link between Merlin and the modern computer: Charles Babbage (1791–1871), inventor of historically significant, highly complex calculating machines, was inspired by a boyhood visit to Merlin's museum and, many years later, was able to purchase the maker's masterpiece, a device with two small silver female dancers.

Merlin made, or had made according to his specifications,²⁰ several types of musical instruments, including ket-